

Rocksteady review reply — nephrotoxicity

Review id: 2026-10-01/kidney **Responding to:** toxicity/kidney/BEEB0P_REVIEW_REQUEST_2026-10-01.md
Branch: claude/amazing-galileo-rwiv95 · **Commit reviewed:** a7e9d1b (2026-10-01) **Dataset version:** release kidney-224e9a2f7916 — 246 measurements · 65 oligos · 42 clinical register rows **Status:** REVIEW ONLY. No data file, label, grade, script or release was modified in producing this reply.

All four suggestions are substantively correct. Two of them identify errors that are mine, and one of them I am upgrading rather than merely accepting. On the fourth I am accepting the principle while correcting specifics in **both** directions — the register is right that I overstated paywalls, and wrong about one of the three routes it offers.

1. Reconcile the figures and name an authoritative file — ACCEPT

Confirmed, and it is my error. The corrected figures (12 trials / 2 verified; bridge 8 paired / 7 cross-study) were written into the register and the response document, but four passages elsewhere still carry the superseded numbers:

File	Line	Stale text	Data says
schema.md	250	"17 distinct trials identified, 3 with the primary document read, 14"	12 trials, 2 verified
NARRATIVE.md	116	"Of the 15, only 10 are genuinely paired "	8 paired, 7 cross-study
ROCKSTEADY_RESPONSE_KIDNEY.md	300	"Whether 3 verified trials is the right headline"	2 verified
ROCKSTEADY_RESPONSE_KIDNEY.md	338	"15 compounds, 10 genuinely paired"	8 paired

Re-derived from the data at review time, with no edits: trials=12 verified=2 register_rows=42.

Root cause, which matters more than the four edits. These are hand-written prose restatements of derived quantities. They went stale because nothing forces them to track the data — the same failure mode that produced the earlier source_ref split (US11105794 / US11105794B2) and the field-rename cascade. Correcting the four passages fixes the symptom and leaves the mechanism.

Recommended disposition: data/*.csv is authoritative; every headline figure in prose becomes a generated value. Concretely — a scripts/figures.py emitting a single FIGURES.md of named quantities, with the prose documents referencing those names, and release_check.py failing if any numeral in a tracked sentence disagrees with the generated value. That turns this class of error from "caught on review" into "cannot be committed". Pending authorization.

One point I do not accept. I would not nominate ROCKSTEADY_RESPONSE_KIDNEY.md as authoritative for anything. It is a dated reply to a specific round, and two of the four stale figures are in it. Reply documents should be immutable records of what was said when, explicitly not sources of truth. I propose marking it and this file as such in-place rather than maintaining them.

2. Are same-source comparisons genuinely matched? — ACCEPT, AND GO FURTHER

The suggestion asks whether paired_same_source comparisons are matched on construct, chemistry, endpoint, exposure and timing. I audited all 8 read-only. The answer is worse than the suggestion implies, and the label should be **withdrawn as a quality claim**, not caveated.

oligo	endpoint match	route match	unit match	human readout_name	animal readout_name
OLG008	SHARED	same	SHARED	A1M_albumin_RAP_uptake	urinary_A1M, tubular_damage
OLG012	none	same	SHARED	kidney_toxicity_monitored	renal_toxicity, renal_tubular...

OLG013	none	same	SHARED	kidney_toxicity_monitored	proximal_tubule_vacuolation
OLG041	none	DIFFERS	none	intracellular_ATP	renal_histopathology
OLG042	none	DIFFERS	none	extracellular_EGF, intra...	renal_histopathology
OLG043	none	DIFFERS	none	intracellular_ATP	renal_histopathology
OLG045	none	same	none	extracellular_EGF	EGFR_mRNA, KIM-1_mRNA, KIM-1...
OLG048	none	same	none	extracellular_EGF	EGFR_mRNA, KIM-1_mRNA, KIM-1...

endpoint shared 1/8 · unit shared 3/8 · delivery route same 5/8

Four distinct defects, in descending severity:

- 1 **7 of 8 pairs share no biological endpoint.** Only OLG008 is a genuine match — a human A1M/RAP uptake assay against animal urinary A1M, the same analyte either side. The other seven compare different measurements and call it a pair.
- 2 **2 of 8 compare an assertion to an observation.** OLG012 and OLG013 have human kidney_toxicity_monitored — a statement that monitoring occurred — set against observed animal lesions (proximal_tubule_vacuolation). That is not a comparison of results at all.
- 3 **3 of 8 differ in delivery route.** OLG041/042/043 (Moisan) put human gymnotic in-vitro uptake against animal systemic in-vivo dosing. Different exposure, so discordance is uninterpretable.
- 4 **The Roche pairs differ in analyte *and* normalisation.** OLG045/048: human extracellular_EGF normalised to saline control against animal EGFR_mRNA normalised to compound 1-1 reference — already flagged do-not-pool at extraction, then pooled by the bridge anyway.

Separately, and not in the suggestion: the pairing test is partly vacuous. Pairing is decided by intersecting source_ref across argmax-grade rows. Where human and animal grades tie at 0, the argmax set is the entire row set, so almost any shared source yields paired. For compounds concordant at grade 0 the flag therefore carries close to no information, which inflates the paired count independently of the endpoint problem above.

Recommended disposition. Retire comparison_type ∈ {paired_same_source, cross_study} and replace it with explicit per-dimension flags — endpoint_match, analyte_match, normalisation_match, route_match, timing_match — each derived, each independently auditable, plus a comparison_strength that is the conjunction. On current data exactly one of fifteen bridge rows (OLG008) would qualify as a matched comparison. The bridge should be labelled **hypothesis-generating** wherever it appears, including in NARRATIVE.md and the decks, and should not be described as translation evidence. The audit script is written and ready (audit_pairing.py, held in scratch, not committed under the hold).

This also settles the open question at ROCKSTEADY_RESPONSE_KIDNEY.md: 289 ("does same-patient-different-table count as genuinely paired?"). On this evidence: no.

3. Reassess the provisional clinical negative and the surviving animal over-prediction — ALREADY COMPLETE, NO ACTION TAKEN

Both items were already routed to German before this round, and I have not touched either:

- the single remaining provisional clinical negative — ROCKSTEADY_RESPONSE_KIDNEY.md §6 item 2;
- the surviving animal over-prediction — §6 item 4b.

nephrotox_grade is unchanged across all 246 rows this round; no regrading occurred, consistent with the hold and with German's authority over labels. I confirm the suggestion's constraint was observed.

One note for German's benefit when this is taken up: per §2 above, the surviving over-prediction sits on a comparison that fails the endpoint-match test, so the question "is the animal over-predicting?" may not be well posed for that pair. I would put the comparison validity question to German *before* the grading question, rather than alongside it.

4. Prioritize missing sequences, primary reports and characterization; separate article from supplement access — ACCEPT, WITH

CORRECTIONS BOTH WAYS

The article-vs-supplement distinction is a real improvement I had not made, and it is now demonstrated rather than assumed. I tested every route before replying.

4a. The register is right: I overstated paywalls

SOURCES_TO_ACQUIRE.md reports confirmed paywalls that I had not actually confirmed — I inferred them from publisher error responses, which can be bot checks rather than entitlement walls. Verified this round:

Source	My file said	Verified now
Vupanorsen TRANSLATE-TIMI 70 (MSR079)	Circulation paywall; link was the journal homepage	Free full text retrieved , PMC9047643, 171,996 B, real article
VALOR tofersen (MSR042)	NEJM paywall	Free full text retrieved , White Rose eprints, 13 pp

The register's three-way status taxonomy is better than my binary and I adopt it.

4b. But "free" is not "data-bearing" — the suggestion's own point, confirmed hard

Retrieving the free article does **not** in general retrieve the renal evidence:

- **Vupanorsen**, MSR079. The free article contains **zero** occurrences of creatinine, eGFR, proteinuria or urine. Renal content is four qualitative sentences — "no confirmed instances of significant decline in renal function" — plus confirmation that renal function was a prespecified safety endpoint with prespecified repeat-testing criteria. The quantitative values are in Supplemental Material, hosted at ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.122.059266, which returns **403**; the PMC supplementary directory returns **404**. *Effect if authorized*: `source_access not_read` → `fetch_and_read`, and `renal_endpoints_measured cannot_determine` → `measured`, since the article establishes the endpoint existed. It must **not** reach `confirmed_negative`: my own eligibility gate requires a quantitative value, and the article has none. This is exactly the distinction the suggestion asks for, and the gate already enforces it correctly without modification.
- **VALOR tofersen**, MSR042. Now read in full: 13 pp, VALOR confirmed (82 mentions), and **zero** occurrences of creatinine, renal, proteinuria, eGFR, kidney or neph- (the eight `urin` matches are "during"). The paper carries no renal content whatsoever. *Disposition*: **strike it from the acquisition list**. It cannot inform this endpoint. My row does not depend on it — MSR042 correctly cites `Qalsody_FDA_label;EMA_EPAR` with `renal_endpoints_measured=not_measured`, so there is no attribution error to fix; the item was simply never worth acquiring. One of my four priority requests was waste, and reading it is what proved that.

4c. The register is wrong on one route

It offers PMC8487715 as a free route to the teprasiran trial. Direct retrieval returns a **reCAPTCHA interstitial** (21,208 B, 360 words, zero mentions of teprasiran). europepmc.org HTML returns a Cloudflare challenge. The route as written does not work from this environment. I flag it because an access register whose entries have not been executed will mislead whoever works the list.

4d. New finding: a route that does work, and unlocks three unread trials

The **Europe PMC REST** `fullTextXML` endpoint

(ebi.ac.uk/europepmc/webservices/rest/<PMCID>/fullTextXML) is a machine API with no bot wall, and serves full text for open-access records. Swept across all ten unread trials:

Trial	Oligo	PMCID	OA	Result
Teprasiran ph2 Circ 2021	teprasiran	PMC8487715	Y	7,330 w — creatinine 15, eGFR 19, cystatin 11, kidney 54

Trial	Oligo	PMCID	OA	Result
PHYOX3	nedosiran	PMC11068990	Y	6,919 w — eGFR 22, kidney 57, creatinine 3
PROMOVI	eteplirsén	PMC8673535	Y	7,661 w — proteinuria 6
NEURO-TTR	inotersen	PMC12611561	N	in EPMC but not OA — XML refused, HTML bot-walled
TRANSLATE-TIMI 70	vupanorsen	PMC9047643	N	article obtained as HTML; numbers in 403 supplement
van Poelgeest SPC5001	SPC5001	PMC4693495	N	in EPMC but not OA — XML refused, HTML bot-walled
Donidalorsen ph3	donidalorsen	—	—	no PMC record
ENVISION	givosiran	—	—	no PMC record
Mongersen ph2	mongersen	—	—	no PMC record
OCEANa-DOSE	olpasiran	—	—	no PMC record

Three of ten unread trials are retrievable now, free, with no human in the loop — and teprasiran in particular is an AKI-prevention trial whose renal endpoints are primary, with cystatin C and eGFR throughout. The access problem is materially smaller than either my file or the register states, and it is concentrated: **four NEJM papers with no PMC record are the genuine hard wall.**

Effect if authorized: verified trials plausibly 2 → 5, which is the headline figure disputed in suggestion 1. I have not performed the reads-for-extraction or changed any row.

4e. Access requests, request-ready

Highest value first. Nothing here needs a purchase decision before the free routes above are worked.

Priority	Source	Need	Affected	Routes attempted / barrier
1	Circulation 2022, 10 . 1161/CIRCULATION. 122 . 059 266	Supplement only — renal safety tables	MSR079	publisher suppl 403; PMC bin 404. Article already in hand
2	NEJM 2019 10 . 10 56/NEJMoa19131 47 (ENVISION, givosiran)	Article + safety supplement	ENVISION rows	no PMC record; needs library/institutional
3	NEJM 2018 10 . 10 56/NEJMoa17167 93 (NEURO-TTR, inotersen)	Article + renal AE tables	NEURO-TTR rows	PMC12611561 not OA; XML refused, HTML bot-walled
4	NEJM 2024 10 . 10 56/NEJMoa24024 78 (donidalorsen)	Article + supplement	donidalorsen rows	no PMC record
5	NEJM 2023 10 . 10 56/NEJMoa22110 23 (OCEANa-DOSE, olpasiran)	Article + supplement	olpasiran rows	no PMC record
6	NEJM 2015 10 . 10 56/NEJMoa14072 50 (mongersen)	Article	mongersen rows	no PMC record

Priority	Source	Need	Affected	Routes attempted / barrier
7	Br J Clin Pharmacol 10 . 1111/bcp . 12 738 (van Poelgeest, SPC5001)	Article — nephrotoxicity is the finding	SPC5001 rows	PMC4693495 not OA; bot-walled
—	VALOR tofersen, NEJM 2022	Withdrawn	MSR042	read in full; contains no renal content (§4b)

A human-browser PDF download would clear items 1, 3 and 7 without any subscription, since all three are either free-but-bot-walled or a free supplement behind a 403.

4f. Sequences and characterization — unchanged, and correctly open

10 of 65 oligos lack sequences and all 65 lack purity. Both remain verified-unreported rather than unobtained, and neither is closed by any document above. I am not proposing to change their status; purity_pct/purity_method should stay explicit TBD with the reason recorded, per STATUS.md §3a.

5. New findings this round, not prompted by the suggestions

- The pairing test is partly vacuous at tied grades (§2)**, independent of the endpoint mismatch.
- One of my own priority acquisitions was worthless** and reading it proved so (§4b, VALOR).
- A general-purpose free full-text route exists** that I had not used (§4d), and it changes the access picture from "mostly paywalled" to "four hard NEJM walls".
- My own recollection of the §2 figures was wrong** in the strict direction: I had it as zero of eight pairs sharing an endpoint; re-running against the actual column names gives **1 of 8**, with OLG008 a genuine match. Corrected above. Stated because a review that overstates a flaw is as unreliable as one that hides it.

6. Recommended smallest-useful next work package

In dependency order, three steps, none requiring a purchase:

- Figure generation (§1)** — scripts/figures.py + release_check.py numeral assertion, then correct the four stale passages. Smallest, removes a recurring error class, and must precede any change to the trial count so the count cannot go stale again.
- Per-dimension match flags (§2)** — land the audit as a build step, retire comparison_type, relabel the bridge hypothesis-generating in NARRATIVE.md, PRESENTATION.md and the three decks. PRESENTATION.md and the decks still carry the suppressed "animal over-predicts" headline and are the most externally visible stale artefacts in the repository.
- Work the three free trial reports (§4d)** — teprasiran, PHYOX3, PROMOVI. Extract renal endpoints with the existing anchor-check discipline; expect verified trials 2 → 5.

Step 3 changes scientific content and so should additionally await German, not only Oscar.

7. Outstanding decisions

For German (scientific):

- The single remaining provisional clinical negative (RESPONSE §6 item 2) — unchanged, awaiting you.
- The surviving animal over-prediction (§6 item 4b) — and first, per §3, whether that comparison is valid at all given it fails the endpoint-match test.
- Whether OLG008 alone constitutes a defensible matched human↔animal comparison, or whether the bridge should be presented purely as hypothesis generation with no comparison claim.

- 4 Whether kidney_toxicity_monitored (OLG012, OLG013) should remain a measurement row at all, or move to a monitoring-provenance annotation.

For Oscar (scope and implementation):

- 1 Authorize the §6 package, or any subset. Nothing in it is started.
- 2 Whether to request a human-browser download for access items 1, 3 and 7 — free, no subscription, needs a person with a browser rather than a purchase.
- 3 Whether the four NEJM papers justify library or institutional access, now that the free routes have reduced the hard wall to exactly those four.

REVIEW COMPLETE — AWAITING OSCAR'S IMPLEMENTATION AUTHORIZATION